AI Social Media Content Detective: Final Project Report

Question: What YouTube content gets Pakistani clothing-brand audiences to engage?
Data snapshot: 2026-10-17 - Report generated: 2026-10-18
Live app: https://pakistani-social-media-content-detective-mahmts48mya2xnxwgttip.streamlit.app
Code: https://github.com/Obaiz1/pakistani-social-media-content-detective

1. Executive summary

Collected 2,204 videos (last 24 months) and 781 comments from 8 official Pakistani clothing-brand channels through the YouTube Data API v3 (no scraping).

After removing low-reliability videos (131) and likely ad-boosted videos (242), 1,845 videos were analysed.

One-at-a-time comparisons suggested that wedding/festive content performs better and that Saturday uploads, menswear and sale content perform worse than the same brand's other videos (small effects).

A mixed-effects model controlling for brand, month and all features together found that 0 of 26 effects stay significant after multiple-comparison correction. Most one-at-a-time differences are largely explained by season and brand mix, so every recommendation is labelled *tentative*.

Data-driven title themes (BERTopic) independently found a festive/luxury theme with the highest relative engagement (festive / luxury / celebrate: 1.34 vs 1.05).

Sentiment: the multilingual XLM-R model (macro-F1 0.82) was chosen over the best LLM (0.87) because the difference was not significant.

Engagement prediction before publishing did not beat a simple median baseline, so it was deferred.

2. Data and compliance

Source: YouTube Data API v3, public data only. Channels pinned by ID (a search had returned a fan channel).

Comment authors are never stored; comment text never leaves the owner's machine (published data has labels only).

Stored API data is refreshed weekly by a GitHub Action and kept as a single latest snapshot (< 30 days).

Not available from the API: shares, saves, reach, impressions, audience demographics.

3. Methods

Engagement rate (ER) = (likes + comments) ÷ views; likes ÷ views where comments are disabled; null if likes are hidden.

Relative engagement = ER ÷ the brand's median ER, which removes brand-size differences.

Likely promoted = like-rate < brand median ÷ 5 and views above the brand's 75th percentile (excluded).

Statistics: medians with 2,000-sample bootstrap 95% CIs; Mann-Whitney U vs all other videos; Cliff's delta effect size; Benjamini-Hochberg correction; groups with n < 20 not tested.

Adjusted model: log(ER) ~ weekday + PKT time band + length + hashtags + campaign tags + emoji, random intercept per brand + month variance component (statsmodels MixedLM); robustness check with brand + month fixed effects (HC3).

NLP: rule-based language ID; 6 sentiment approaches benchmarked on 250 hand-labelled real comments.

Topics: BERTopic on titles (multilingual MiniLM embeddings, UMAP, HDBSCAN, reduced to 12 themes).

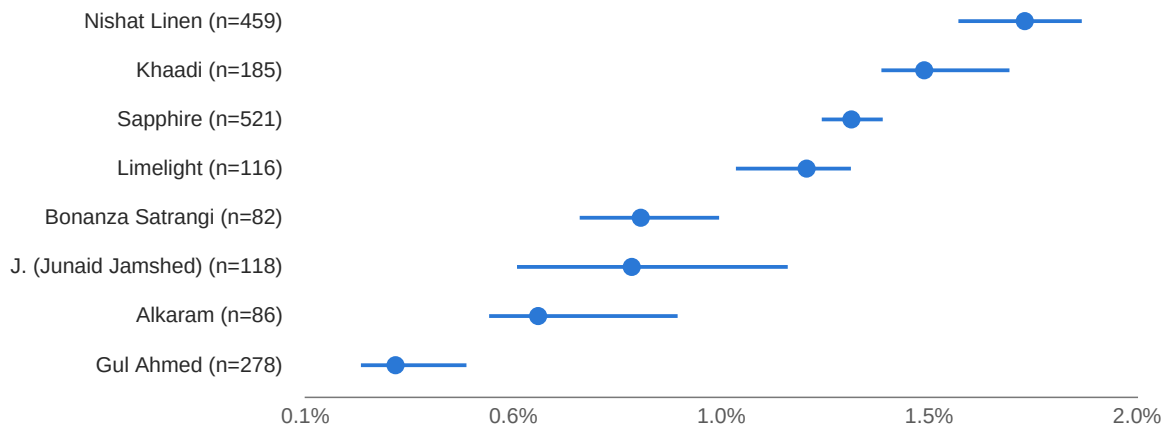
Thumbnails: colour statistics for every video; vision-LLM labels (NVIDIA Nemotron) for a random sample, spot-checked against manual labels.

Prediction: chronological split (newest 20% as test), pre-publication features only, two baselines.

4. Results

4.1 Engagement by brand (median ER, 95% CI)

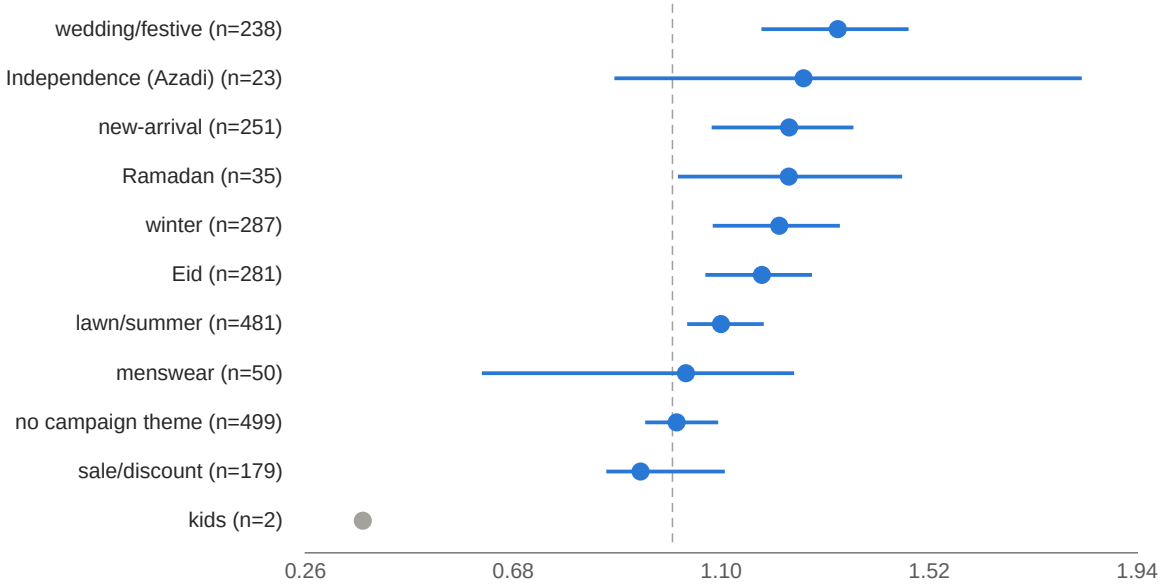
Median engagement rate by brand



Nishat Linen: median 1.71% [1.57%, 1.83%], n = 459, medium effect, adj. p = 0.000
Khaadi: median 1.49% [1.39%, 1.67%], n = 185, small effect, adj. p = 0.000
Sapphire: median 1.32% [1.26%, 1.39%], n = 521, negligible effect, adj. p = 0.000
Limelight: median 1.22% [1.07%, 1.32%], n = 116, negligible effect, adj. p = 0.426
Bonanza Satrangi: median 0.85% [0.72%, 1.02%], n = 82, small effect, adj. p = 0.000
J. (Junaid Jamshed): median 0.83% [0.58%, 1.18%], n = 118, small effect, adj. p = 0.000
Alkaram: median 0.62% [0.52%, 0.93%], n = 86, medium effect, adj. p = 0.000
Gul Ahmed: median 0.30% [0.23%, 0.46%], n = 278, large effect, adj. p = 0.000

4.2 Content and timing (relative engagement, one feature at a time)

Relative engagement by campaign (1.0 = brand median)



Campaign

wedding/festive: median 1.34 [1.18, 1.47], n = 238, small effect, adj. p = 0.001
Independence (Azadi): median 1.27 [0.89, 1.83], n = 23, negligible effect, adj. p = 0.896
new-arrival: median 1.24 [1.08, 1.36], n = 251, negligible effect, adj. p = 0.387
Ramadan: median 1.24 [1.01, 1.46], n = 35, negligible effect, adj. p = 0.706
winter: median 1.22 [1.09, 1.34], n = 287, negligible effect, adj. p = 0.449
Eid: median 1.18 [1.07, 1.28], n = 281, negligible effect, adj. p = 0.298
lawn/summer: median 1.10 [1.03, 1.18], n = 481, negligible effect, adj. p = 0.896
menswear: median 1.03 [0.62, 1.24], n = 50, small effect, adj. p = 0.045
no campaign theme: median 1.01 [0.95, 1.09], n = 499, negligible effect, adj. p = 0.007
sale/discount: median 0.94 [0.87, 1.10], n = 179, negligible effect, adj. p = 0.007
kids: median 0.37 (n < 20, not tested), n = 2

Video length

0-15s: median 1.05 [0.98, 1.12], n = 589, negligible effect, adj. p = 0.010
16-30s: median 1.11 [1.05, 1.18], n = 834, negligible effect, adj. p = 0.078

31-60s: median 1.18 [1.09, 1.28], n = 355, negligible effect, adj. p = 0.404
61-180s: median 1.01 [0.78, 1.75], n = 63, negligible effect, adj. p = 0.932
>180s: median 1.17 (n < 20, not tested), n = 4

PKT time band

Night (00-06): median 1.65 [0.95, 2.59], n = 27, small effect, adj. p = 0.111
Morning (06-12): median 1.20 [1.10, 1.31], n = 360, negligible effect, adj. p = 0.042
Afternoon (12-16): median 1.12 [1.03, 1.23], n = 556, negligible effect, adj. p = 0.465
Evening (16-20): median 1.04 [1.00, 1.10], n = 578, negligible effect, adj. p = 0.030
Prime time (20-24): median 1.07 [1.01, 1.20], n = 324, negligible effect, adj. p = 0.465

Weekday

Monday: median 1.17 [1.03, 1.30], n = 349, negligible effect, adj. p = 0.437
Tuesday: median 1.15 [1.09, 1.25], n = 306, negligible effect, adj. p = 0.437
Wednesday: median 1.13 [1.02, 1.26], n = 272, negligible effect, adj. p = 0.437
Thursday: median 1.05 [0.97, 1.17], n = 355, negligible effect, adj. p = 0.437
Friday: median 1.10 [1.04, 1.24], n = 276, negligible effect, adj. p = 0.437
Saturday: median 0.93 [0.82, 1.02], n = 183, small effect, adj. p = 0.006
Sunday: median 1.03 [0.94, 1.18], n = 104, negligible effect, adj. p = 0.437

4.3 Adjusted effects (mixed-effects model)

n = 1,830 videos, 8 brands, 25 months. % change in ER vs the reference level, with 95% CI and BH-adjusted p:

Campaign: independence: +25% [-24%, +105%], adj. p = 0.64
Emoji in title: +24% [-25%, +105%], adj. p = 0.64
Weekday: Wednesday (vs Monday): +22% [+5%, +42%], adj. p = 0.10
Campaign: lawn/summer: +19% [+4%, +37%], adj. p = 0.10
Weekday: Sunday (vs Monday): +14% [-8%, +41%], adj. p = 0.44
Weekday: Friday (vs Monday): +14% [-2%, +33%], adj. p = 0.39
Campaign: eid: +14% [-4%, +35%], adj. p = 0.39
Campaign: wedding/festive: +12% [-3%, +30%], adj. p = 0.39
Time band: Morning (06-12) (vs Afternoon (12-16)): +12% [-2%, +28%], adj. p = 0.39
Weekday: Tuesday (vs Monday): +11% [-4%, +29%], adj. p = 0.41
Campaign: new/arrivals: +10% [-5%, +27%], adj. p = 0.44
Hashtags: 1-2 (vs 0): +6% [-30%, +63%], adj. p = 0.87
Hashtags: 3-5 (vs 0): +4% [-13%, +25%], adj. p = 0.83
Time band: Prime time (20-24) (vs Afternoon (12-16)): +4% [-11%, +21%], adj. p = 0.83
Campaign: winter: +4% [-13%, +24%], adj. p = 0.83
Time band: Night (00-06) (vs Afternoon (12-16)): +4% [-28%, +50%], adj. p = 0.89
Weekday: Thursday (vs Monday): +2% [-12%, +18%], adj. p = 0.87
Length: 31-60s (vs 16-30s): -1% [-13%, +13%], adj. p = 0.92
Hashtags: 6+ (vs 0): -2% [-20%, +21%], adj. p = 0.89
Length: 0-15s (vs 16-30s): -6% [-15%, +5%], adj. p = 0.55
Time band: Evening (16-20) (vs Afternoon (12-16)): -8% [-19%, +4%], adj. p = 0.44
Length: 61-180s (vs 16-30s): -8% [-29%, +18%], adj. p = 0.72
Campaign: ramadan: -12% [-38%, +26%], adj. p = 0.72
Weekday: Saturday (vs Monday): -13% [-27%, +4%], adj. p = 0.39
Campaign: sale: -19% [-33%, -3%], adj. p = 0.15
Campaign: men: -31% [-47%, -9%], adj. p = 0.10

4.4 Title themes (BERTopic)

459 videos had no clear theme (outliers, excluded).

festive / luxury / celebrate: median 1.34 [1.15, 1.48], n = 264, small effect, adj. p = 0.000
unstitched / summer: median 1.25 [0.83, 1.59], n = 47, negligible effect, adj. p = 0.892
intermix / unstitched: median 1.21 [0.88, 1.28], n = 26, negligible effect, adj. p = 0.892
sale / mothers / tribute: median 1.20 [1.01, 1.55], n = 129, negligible effect, adj. p = 0.677
signature / premium / elegance: median 1.16 [1.02, 1.31], n = 53, negligible effect, adj. p = 0.892
west / azadi / eid: median 1.12 [0.86, 1.33], n = 113, negligible effect, adj. p = 0.892
lawn / spring / summer: median 1.09 [0.97, 1.22], n = 237, negligible effect, adj. p = 0.892
winter / fall / unstitched: median 1.08 [0.95, 1.22], n = 171, negligible effect, adj. p = 0.706
ready / eid / bloom: median 1.02 [0.97, 1.11], n = 259, negligible effect, adj. p = 0.131

eid / pret / buy: median 0.91 [0.80, 1.15], n = 68, small effect, adj. p = 0.099
centre / city / experience: median 0.90 [0.74, 1.31], n = 25, small effect, adj. p = 0.496
flat / pret / sale: median 0.84 [0.75, 0.92], n = 62, small effect, adj. p = 0.020

4.6 Comment sentiment

Overall: 49% positive, 33% neutral, 18% negative. Net sentiment (share positive - share negative) by brand:

Alkaram: +0.50 [+0.33, +0.65], n = 82
Khaadi: +0.45 [+0.26, +0.62], n = 53
Nishat Linen: +0.43 [+0.34, +0.54], n = 196
Limelight: +0.40 [+0.24, +0.56], n = 70
Sapphire: +0.19 [+0.09, +0.29], n = 243
Gul Ahmed: +0.12 [-0.03, +0.26], n = 116
J. (Junaid Jamshed): +0.05 [-0.29, +0.38], n = 21

4.7 Sentiment model benchmark (macro-F1, n = 250)

nvidia:nvidia/nemotron-3-super-120b-a12b: 0.869
cardiffnlp/twitter-xlm-roberta-base-sentiment: 0.819
routed: EN->cardiff-roberta, RU/mixed->roman-urdu-xlmr: 0.786
Khubaib01/roman-urdu-sentiment-xlm-r: 0.763
cardiffnlp/twitter-roberta-base-sentiment-latest: 0.721
vader: 0.602

Label status: PROVISIONAL: labels are Claude drafts, not yet verified by the owner.

4.8 Engagement prediction

baseline: global median: MAE(log) 0.609
baseline: brand median: MAE(log) 0.699
ridge: MAE(log) 0.679
gradient boosting: MAE(log) 0.682

No model beat the stronger baseline, so prediction was deferred.

5. Recommendations (to A/B test, not guarantees)

Next post to test [tentative (not confirmed by the adjusted model), n = 238]
What: A wedding/festive video. De-prioritise: menswear content. When: Any day except Saturday (PKT); no reliable best-time signal.
Evidence: Wedding/festive content: median relative engagement 1.34 (95% CI 1.18-1.47) vs 1.10 for all other videos (1.00 = a brand's typical video). Cliff's delta +0.15 (small effect), BH-adjusted p = 0.001.
Adjusted check: Campaign: wedding/festive: +12% (95% CI -3% to +30%), BH-adjusted p = 0.39 -> NOT confirmed after adjusting for brand, month and other features.
Test: Run it as a 4-week A/B: half of comparable uploads follow this plan, half follow the usual plan; compare median engagement after 14 days per video (>= 20 videos per arm).

Use less: menswear content [tentative (not confirmed by the adjusted model), n = 50]
What: De-prioritise menswear content; test alternatives. When: Keep your usual schedule (see the timing cards).
Evidence: Menswear content: median relative engagement 1.03 (95% CI 0.62-1.24) vs 1.13 for all other videos (1.00 = a brand's typical video). Cliff's delta -0.19 (small effect), BH-adjusted p = 0.045.
Adjusted check: Campaign: men: -31% (95% CI -47% to -9%), BH-adjusted p = 0.10 -> NOT confirmed after adjusting for brand, month and other features.
Test: Publish 10-20 menswear content and 10-20 otherwise-similar videos in the same weeks and time slots. Compare median engagement (likes + comments ÷ views) after 14 days; keep ad spend equal or exclude boosted posts.

Try more: wedding/festive content [tentative (not confirmed by the adjusted model), n = 238]
What: Test more wedding/festive content. When: Keep your usual schedule (see the timing cards).
Evidence: Wedding/festive content: median relative engagement 1.34 (95% CI 1.18-1.47) vs 1.10 for all other videos (1.00 = a brand's typical video). Cliff's delta +0.15 (small effect), BH-adjusted p = 0.001.
Adjusted check: Campaign: wedding/festive: +12% (95% CI -3% to +30%), BH-adjusted p = 0.39 -> NOT confirmed after adjusting for brand, month and other features.
Test: Publish 10-20 wedding/festive content and 10-20 otherwise-similar videos in the same weeks and time slots. Compare median engagement (likes + comments ÷ views) after 14 days; keep ad spend equal or exclude boosted posts.

Use less: posting on Saturday [tentative (not confirmed by the adjusted model), n = 183]
What: Move some uploads away from Saturday. When: Avoid Saturday (PKT) until tested.
Evidence: Posting on Saturday: median relative engagement 0.93 (95% CI 0.82-1.02) vs 1.12 for all other videos (1.00 = a brand's typical video). Cliff's delta -0.15 (small effect), BH-adjusted p = 0.006.
Adjusted check: Weekday: Saturday (vs Monday): -13% (95% CI -27% to +4%), BH-adjusted p = 0.39 -> NOT confirmed after adjusting for brand, month and other features.
Test: For 4 weeks, alternate comparable videos between Saturday and your usual slot (same campaign, length and hashtags). Compare median engagement after 14 days per video; aim for at least 20 videos per arm.

6. Limitations

- Pakistan-related brand content: the audience is not verified as Pakistani (no audience-location data in the API).
- Correlation, not causation: posting choices and outcomes share hidden causes (budget, season, product).
- View counts can include paid ads; 'likely promoted' videos are excluded by default (heuristic flag).
- Groups with n < 20 are not tested. Effect sizes are mostly small or negligible.
- Public YouTube metrics only: shares, saves, reach and impressions are not available.
- Only 5 Urdu-script comments: Urdu-script sentiment could not be evaluated.
- BERTopic themes vary slightly between runs; vision labels for text overlay/close-up are low-agreement.

7. Engineering

- Modular Python package (src/), Streamlit dashboard (10 pages), 77 automated tests, 28 architecture decision records.
- Weekly GitHub Action -> private data repo (deploy key) -> Streamlit Community Cloud (read-only token).
- Secrets only in .env / Streamlit secrets; a pre-commit check blocks any secret value from git.

8. Process

Built in 15 gated phases (problem framing -> requirements -> data audit -> architecture -> foundation -> pipeline -> analytics MVP -> NLP -> timing -> ML -> recommendations -> dashboard & exports -> delivery), with the project owner approving every gate. Full history: DECISIONS.md, PROGRESS.md, SESSION_HANDOFF.md.

9. AI-use disclosure

Built with AI assistance (Claude Code) under the owner's direction. Sentiment benchmark labels were drafted by Claude for owner verification; the NVIDIA API Catalog was used for an LLM sentiment baseline, recommendation wording and thumbnail labels.